MANAS SHUKLA

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AUTOMATION & MACHINE LEARNING ENGINEER

Passionate about bridging control engineering with AI to drive scalable, intelligent automation solutions.

A decade of cross-functional experience in industrial automation, robotics, and ML/AI. Skilled in developing intelligent control architectures, integrating machine learning models into real-world manufacturing systems, and building high-efficiency pipelines for autonomous decision-making. Proven ability to lead projects across domains, from PLC programming and HMI design to ML deployment and algorithm optimization. My solutions have been deployed in global production lines, enhancing throughput, quality control, and system scalability.

TECHNICAL SKILLS

Control Systems (PLCs):	Beckhoff | Siemens | Allen Bradley | Mitsubishi
Automation Tools:	TwinCAT | TIA Portal | RSLogix | GXWorks | Connected Components Workbench (CCW) | Factory I/O | SCADA | HMI (WinForms, Ignition) | MES
Programming Languages:	Python | C | C++ | C# | R | MATLAB | LabVIEW | IEC 61131-3 (Ladder Logic, Structured Text)
Databases:	MS SQL | PostgreSQL
ML/CV Tools:	TensorFlow | PyTorch | Keras | OpenCV | Scikit-learn | HOG | SVM | CNN | EKF | AWS EC2/S3
Tools & Platforms:	ROS | Gazebo | Simulink | SolidWorks | NX | Jupyter Notebook | Anaconda | GitHub | SVN | Jira

EXPERIENCE

Sensata Technologies, Attleboro, MA, USA — **Jan 2022 – Mar 2025**

Senior Systems Engineer, Mechanization Team

Led end-to-end design and integration of complex, multi-station automation systems used globally for manufacturing and testing Sensata products, including calibrators, characterizers, final functional testers, and other high-precision systems.

- Developed control logic and operational sequences across various PLC platforms, enhancing code reusability by 80% and decreasing errors by 20% through a modular architecture design philosophy..
- Designed HMI interfaces using WinForms and TwinCAT; built SCADA dashboards for intuitive user control.
- Interfaced automation equipment with MES using Python-SQL pipelines and a modular C# .NET API for Beckhoff PLCs, improving traceability, analytics, and data transfer speed by 5x.
- Configured servo drives, actuators, robots, and VFDs through Modbus, EtherCAT, and Ethernet/IP protocols.
- Configured safety systems across robotic gantries, conveyors, and pneumatic systems.
- Managed project timelines, scope, and budgets while collaborating with global suppliers for successful deployment.
- Conducted GRR and ANOVA studies for process validation and supported with machine qualifications.

Universal Logic Inc., Charlotte, NC, USA — **May 2018 – Jan 2022**

Machine Learning Engineer, AI Labs

Developed AI-powered robotic systems for dynamic material handling applications, integrating vision, motion control, and real-time decision-making to enhance automation in supply chain operations.

- Deployed various machine learning and deep learning algorithms to enable robots to adapt to variable SKUs and unstructured environments, improving pick-and-place accuracy and efficiency.

- Integrated Neocortex AI software with robotic hardware to facilitate real-time perception and manipulation, reducing manual intervention in sorting and palletizing tasks.
- Designed databases and web dashboards to streamline data visualization and model monitoring for the end user.
- Developed modular software components, enhancing the flexibility and maintainability of robotic applications across diverse client requirements.
- Researched emerging ML techniques to continuously refine algorithm performance and support product innovation.

University of Pennsylvania, Philadelphia, PA, USA **Jun 2016 – May 2018**

Research Assistant, Real-Time & Embedded Systems Lab (mLab) Jun 2017 – May 2018

Engineered autonomous vehicles' navigation algorithms for research and simulation environments.

- Built control algorithms to generate custom waypoints for multi-agent coordination and obstacle avoidance.
- Designed a simulation suite with ROS/Gazebo and CAD models for trajectory testing; open-sourced on GitHub (20+ stars) and used by academic collaborators.

Research Assistant, Sinno Research Group, Vagelos Laboratory Jun 2016 – May 2017

Developed high-performance simulations in FORTRAN and C++ to study platelet behavior in complex fluid dynamics.

- Built computational models based on the Lattice Boltzmann Method to simulate platelet motion in blood flow.
- Evaluated different solver algorithms by assessing scalability, numerical stability, and error margins.

Quest-Global Inc., Bangalore, India & Bristol, United Kingdom **Jun 2011 – Oct 2013**

Design Automation Engineer, Embedded Systems Team

Delivered parametric CAD automation and cost-optimization solutions for aeroplane engine design at Rolls-Royce.

- Utilized Siemens NX Knowledge Fusion to automate parametric modeling workflows for turbine blade components.
- Created reusable features and user-friendly GUIs to streamline design modifications and reduce design time by 40%.
- Led UK on-site deployment, led customer reviews, delivered 3 projects on time, and co-developed geometry optimization workflows using Simulia Isight, achieving a 100% customer satisfaction (CSat) score.
- Authored technical content viewed over 500 times; earned company-wide performance and on-time delivery awards.

EDUCATION

Master of Science (MS), Mechanical Engineering, **University of Pennsylvania,** Philadelphia, PA, USA

Bachelor of Technology (BTech), Mechanical Engineering, **Vellore Institute of Technology University,** Vellore, TN, India

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SELECTED PROJECTS

Path Planning for Autonomous Driving:

Developed a highway navigation algorithm for autonomous vehicles using behavioral planning, cost functions, and spline interpolation for safe, smooth trajectories.

PID Controller for Steering Control:

Designed and tuned a PID controller to minimize cross-track error and maintain stable steering in car driving simulations.

Sensor Fusion with Extended Kalman Filter:

Implemented EKF to fuse Lidar and Radar data, enabling real-time state estimation validated against ground-truth RMSE.

Vehicle Detection Using Machine Learning:

Built a vehicle detection pipeline using HOG features, SVM, and color transformations, integrated into a real-time video stream.

For more information and code samples, please visit shuklam.info or explore selected repositories at github.com/smanas20.